The complete deficit-four terminal classification

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For a positive exponent word $w = (a_0, \dots, a_{r-1})$, put

$$A_0 := 0, \qquad A_k := \sum_{i=0}^{k-1} a_i, \qquad C_0 := 0, \qquad C_{k+1} := 3C_k + 2^{A_k}, \tag{1}$$

and define

$$D(w) := C_r + 2^{A_r}. \tag{2}$$

The word is a strict barrier word if

$$2^{A_k} < 3^k \qquad (1 \le k \le r). \tag{3}$$

We use the accelerated maps

$$U_+(n) := \frac{3n+1}{2^{v_2(3n+1)}}, \qquad U_-(m) := \frac{3m-1}{2^{v_2(3m-1)}} \tag{4}$$

on positive odd integers.

Lemma 1 (terminal reconstruction and finite seed bound)**.** *Write a strict barrier word as $w = uv$, where $|u| = d$ and $|v| = \ell$. If*

$$3^\ell \mid D(w), \tag{5}$$

then v is the exact exponent word of a positive U_--trajectory from

$$m_0 := \frac{D(v)}{3^\ell} \tag{6}$$

to 1*. Until its first hit of* 1*, this trajectory obeys*

$$\frac{3^k}{2^{A_k(v)}} > c_u := \frac{2^{A(u)}}{3^d}. \tag{7}$$

Consequently,

$$\boxed{m_0 < c_u^{-1} e^{10.789} < \frac{3^{d+11}}{2^{A(u)}}.} \tag{8}$$

Proof. The concatenation identity is

$$D(uv) = 3^{\ell} C(u) + 2^{A(u)} D(v). \tag{9}$$

Since 2 is a unit modulo 3, condition (5) implies $3^{\ell} \mid D(v)$. We verify the terminal reconstruction directly. For $0 \le k \le \ell$, let $v_{\ge k}$ be the suffix beginning at k and put

$$m_k := \frac{D(v_{\ge k})}{3^{\ell-k}}. \tag{10}$$

Equation (9), applied at the cut after the first k letters, shows inductively that every m_k is an integer. Each $D(v_{\ge k})$ is odd, so every m_k is odd. Since $D(\varnothing) = 1$, we have $m_{\ell} = 1$. The one-letter form of (9) gives

$$3m_k - 1 = 2^{a_k} m_{k+1}. \tag{11}$$

Thus the powers of 2 are exact and the asserted U_--trajectory exists.

For every prefix of v, the barrier inequality for the corresponding prefix of uv gives (7). Truncate the negative trajectory at its first hit of 1. It is simple before that hit. The proved universal reciprocal-mass bound for finite simple negative shortcut trajectories is

$$\sum_i \frac{1}{3m_i - 1} < 10.789. \tag{12}$$

The exact product formula and (7) therefore give

$$\log m_0 < -\log c_u + \sum_i \frac{1}{3m_i - 1} < -\log c_u + 10.789.$$

Finally, $e^{10.789} < 3^{11}$, which proves (8). □

Theorem 2 (complete deficit-four classification)**.** *Let w be a strict barrier word of length $r \ge 4$. Define the stem set*

$$\mathcal{S}_4 := \{\varnothing, 12, 112, 1112, 1113, 1122, 1212, 11113\}. \tag{13}$$

Then

$$\boxed{3^{r-4} \mid D(w) \quad \Longleftrightarrow \quad w = s1^{r-|s|} \text{ for some } s \in \mathcal{S}_4 \text{ with } |s| \le r.} \tag{14}$$

Proof. The length-four strict barrier prefixes are exactly

$$1111, \quad 1112, \quad 1113, \quad 1121, \quad 1122, \quad 1211, \quad 1212. \tag{15}$$

Indeed, (3) first forces $a_0 = 1$. At lengths two, three, and four it forces $A_2 \le 3$, $A_3 \le 4$, and $A_4 \le 6$, respectively, which gives (15) by immediate enumeration. Grouped by total exponent, the prefixes are

$A(u)$	c_u	u
4	16/81	1111
5	32/81	1112, 1121, 1211
6	64/81	1113, 1122, 1212.

(16)

Lemma 1 bounds the possible negative-shadow seeds by

$A(u)$	$m_0 < 3^{15}/2^{A(u)}$
4	896806.6875
5	448403.34375
6	224201.671875.

(17)

Two independent exact scans classify every odd seed below these three bounds. The accelerated scan extracts $v_2(3m-1)$ directly. The shortcut scan applies one division by 2 at a time and never extracts a valuation. Both scans stop only at 1, at a repeated state, or when (7) fails. Their common partitions are

c_u	hit 1	floor failure	cycle	first-hit seeds and words
16/81	2	448381	20	$(1,\varnothing),(3,(3))$
32/81	1	224188	13	$(1,\varnothing)$
64/81	1	112097	3	$(1,\varnothing)$.

(18)

No step cap is reached. The longest accelerated resolutions have lengths 139, 128, and 93; the independent shortcut resolutions have lengths 224, 207, and 149. These computations are checked by the exact certificates

`sync/deficit_four_scan.js`, `sync/deficit_four_shortcut_scan.js`.

After a negative trajectory reaches 1, determinism gives only exponent-one steps. Combining (15), (16), and (18) therefore gives precisely the stem list (13): the seed 3 after the prefix 1111 contributes the additional stem 11113, and every other accepted suffix is all ones.

For the reverse implication, $D(v1) = 3D(v)$, and direct calculation gives

s	$\varnothing$	12	112	1112	1113	1122	1212	11113
$D(s)$	1	13	35	97	129	137	149	339
$v_3(D(s))$	0	0	0	0	1	0	0	1.

(19)

Thus $v_3(D(s)) \ge |s| - 4$ for every stem in (13), and appending $r - |s|$ ones proves $3^{r-4} \mid D(w)$. □

Corollary 3 (fixed-start deficit-four horizon)**.** *Fix a positive odd block start x. For a stem $s \in \mathcal{S}_4$, say that s matches x if the first $|s|$ accelerated exponents of x equal s. Define*

$$R_4(x) := \max_{\substack{s\in\mathcal{S}_4 \\ s \text{ matches } x}} \Big(|s| + v_2\big(U_+^{|s|}(x)+1\big) - 1 \Big). \tag{20}$$

The empty stem always matches, so this maximum is defined. If $r \geq 4$, the first r exponents from x form a strict barrier word, and $r > R_4(x)$, then

$$\boxed{v_3\big(U_+^r(x)+1\big) \leq r-5.} \tag{21}$$

Consequently, to exclude a normalized divergent orbit it is sufficient to find a tail-record start x_j and a length r such that

$$r \geq 4, \qquad r > R_4(x_j), \qquad v_3(x_{j+r}+1) \geq r-4. \tag{22}$$

Proof. Suppose that the left side of (21) were at least $r-4$. The positive affine identity gives

$$2^{A_r}\big(U_+^r(x)+1\big) = 3^r x + D(w), \tag{23}$$

so $3^{r-4} \mid D(w)$. Theorem 2 gives $w = s1^{r-|s|}$ for one matching stem s. A run of L exponent-one steps from a positive odd value y has length at most $v_2(y+1)-1$. Hence

$$r - |s| \leq v_2\big(U_+^{|s|}(x)+1\big) - 1,$$

which says $r \leq R_4(x)$, a contradiction. This proves (21).

At a tail-record start every finite future block is a strict barrier block. Thus (22) contradicts (21), which proves the final assertion. □